

Assignment No : 12

Title : Deploy and run the project in AWS without using the port.

Step 1: Create an EC2 instance as created in Asg 11.

aws

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EC2

Instances

Launch an instance

It seems like you may be new to launching instances in EC2. Take a walkthrough to learn about EC2, how to launch instances and about best practices

Do not show me this message again

Take a walkthrough

Launch an instance

Info

Amazon EC2 allows you to create virtual machines, or instances, that run on the AWS Cloud. Quickly get started by following the simple steps below.

Name and tags

Info

Name

inst1

Add additional tags

Application and OS Images (Amazon Machine Image)

Info

An AMI is a template that contains the software configuration (operating system, application server, and applications) required to launch your instance. Search or Browse for AMIs if you don't see what you are looking for below

Search our full catalog including 1000s of application and OS images

Recents

Quick Start

Amazon Linux

macOS

Ubuntu

Windows

Red Hat

SUSE Linux

Debian

Browse more AMIs

Summary

Number of instances

Info

1

Software Image (AMI)

Amazon Linux 2023 AMI 2023.7.2...read more

ami-0f1dccc63b6b9a6438

Virtual server type (instance type)

t2.micro

Firewall (security group)

New security group

Storage (volumes)

1 volume(s) - 8 GiB

Free tier: In your first year of opening an AWS account, you get 750 hours per month of t2.micro instance usage for t3.micro where

Application and OS Images (Amazon Machine Image)

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Recents

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Debian

Browse more AMIs

Amazon Machine Image (AMI)

Ubuntu Server 24.04 LTS (HVM), SSD Volume Type

ami-0e35ddab05955cf57 (64-bit (x86)) / ami-0429d68a1cd41ca80 (64-bit (Arm))

Virtualization: hvm

ENA enabled: true

Root device type: ebs

Free tier eligible

Description

Ubuntu Server 24.04 LTS (HVM),EBS General Purpose (SSD) Volume Type. Support available from Canonical (http://www.ubuntu.com/cloud/services).

Canonical, Ubuntu, 24.04, amd64 noble image

Summary

Number of instances

Info

1

Software Image (AMI)

Canonical, Ubuntu, 24.04, amd6...read more

ami-0e35ddab05955cf57

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Cancel

Launch instance

Preview code

Key pair (login)

Info

You can use a key pair to securely connect to your instance. Ensure that you have access to the selected key pair before you launch the instance.

Key pair name - required

key-S1

Create new key pair

Network settings

Info

Network

Info

vpc-0e2181ce3777a313c

Subnet

Info

No preference (Default subnet in any availability zone)

Auto-assign public IP

Info

Enable

Additional charges apply when outside of free tier allowance

Firewall (security groups)

Info

A security group is a set of firewall rules that control the traffic for your instance. Add rules to allow specific traffic to reach your instance.

Create security group

Select existing security group

We'll create a new security group called 'launch-wizard-1' with the following rules:

Allow SSH traffic from

Helps you connect to your instance

Anywhere

Summary

Number of instances

Info

1

Software Image (AMI)

Canonical, Ubuntu, 24.04, amd6...read more

ami-0e35ddab05955cf57

Virtual server type (instance type)

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Firewall (security group)

New security group

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Cancel

Launch instance

Preview code

aws

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EC2

Instances

Launch an instance

▼ Network settings

Info

Edit

Network

Info

vpc-0e2181ce3777a313c

Subnet

Info

No preference (Default subnet in any availability zone)

Auto-assign public IP

Info

Enable

Additional charges apply when outside of free tier allowance

Firewall (security groups)

Info

A security group is a set of firewall rules that control the traffic for your instance. Add rules to allow specific traffic to reach your instance.

Create security group

Select existing security group

Common security groups

Info

Select security groups

secure51 sg-07c04a05d7396992f VPC: vpc-0e2181ce3777a313c

Compare security group rules

Security groups that you add or remove here will be added to or removed from all your network interfaces.

▼ Configure storage

Info

Advanced

1x 8 GiB gp3 Root volume, 3000 IOPS, Not encrypted

▼ Summary

Number of instances

1

Software Image (AMI)

Canonical, Ubuntu, 24.04, amd64...read more

ami-0e35ddab05955cf57

Virtual server type (instance type)

t2.micro

Firewall (security group)

secure51

Storage (volumes)

1 volume(s) - 8 GiB

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Cancel

Launch instance

Preview code

aws

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EC2

Instances

Launch an instance

Success

Successfully initiated launch of instance (i-01b4c2afa3036c9cf)

Step 2: Connect the instance to the server.

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EC2

Instances

i-01b4c2afa3036c9cf

EC2

Dashboard

EC2 Global View

Events

▼ Instances

Instances

Instance Types

Launch Templates

Spot Requests

Savings Plans

Reserved Instances

Dedicated Hosts

Capacity Reservations

▼ Images

AMIs

AMI Catalog

▼ Elastic Block Store

Volumes

Snapshots

Instance summary for i-01b4c2afa3036c9cf (inst1)

Info

Updated less than a minute ago

Instance ID

i-01b4c2afa3036c9cf

IPv6 address

-

Hostname type

IP name: ip-172-31-11-208.ap-south-1.compute.internal

Answer private resource DNS name

IPv4 (A)

Auto-assigned IP address

13.233.131.21 [Public IP]

IAM Role

-

IMDSv2

Required

Public IPv4 address

13.233.131.21 | open address

Instance state

Running

Private IP DNS name (IPv4 only)

ip-172-31-11-208.ap-south-1.compute.internal

Instance type

t2.micro

VPC ID

vpc-0e2181ce3777a313c

Subnet ID

subnet-0e06b1c09b402835f

Instance ARN

arn:aws:ec2:ap-south-1:586794457897:instance/i-01b

Private IPv4 addresses

172.31.11.208

Public IPv4 DNS

ec2-13-233-131-21.ap-south-1.compute.amazonaws.com | open address

Elastic IP addresses

-

AWS Compute Optimizer finding

Opt-in to AWS Compute Optimizer for recommendation s. | Learn more

Auto Scaling Group name

-

Managed

false

Connect

Instance state

Actions

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EC2 > Instances > i-01b4c2afa3036c9cf > Connect to instance

Connect to instance Info

Connect to your instance i-01b4c2afa3036c9cf (inst1) using any of these options

EC2 Instance Connect
Session Manager
SSH client
EC2 serial console

Instance ID

i-01b4c2afa3036c9cf (inst1)

Connection Type

☒ **Connect using EC2 Instance Connect**
Connect using the EC2 Instance Connect browser-based client, with a public IPv4 or IPv6 address.

☐ **Connect using EC2 Instance Connect Endpoint**
Connect using the EC2 Instance Connect browser-based client, with a private IPv4 address and a VPC endpoint.

Public IPv4 address

13.233.131.21

IPv6 address

-

Username

Enter the username defined in the AMI used to launch the instance. If you didn't define a custom username, use the default username, ubuntu.

ubuntu

Note: In most cases, the default username, ubuntu, is correct. However, read your AMI usage instructions to check if the AMI owner has changed the default AMI username.

Cancel **Connect**

Step 3: ubuntu > cd / > ls > cd etc > ls > cd nginx > ls > cd sites-available > ls > sudo chmod 777 default > nano default

and write the following code in location

“{

```
proxy_pass http://localhost:4000;
proxy_http_version 1.1;
proxy_set_header Upgrade $http_upgrade;
proxy_set_header connection 'Upgrade';
proxy_set_header Host $host;
proxy_cache_bypass $http_upgrade;
```

}” and save the default file.

```
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```

```
ubuntu@ip-172-31-11-208:~$ cd /
ubuntu@ip-172-31-11-208:/$ ls
Prasun_aws  bin.usr-is-merged  dev  home  lib.usr-is-merged  lost+found  mnt  proc  run  sbin.usr-is-merged  srv  var
bin  boot  etc  lib  lib64  media  opt  root  sbin  snap  sys  usr

ubuntu@ip-172-31-11-208:/$ cd etc
ubuntu@ip-172-31-11-208:/etc$ ls
ModemManager  cron.hourly  groff  ld.so.conf  modprobe.d  perl  screenrc  sysstat
PackageKit  cron.monthly  group  ld.so.conf.d  modules  pki  security  systemd
X11  cron.weekly  group-  ldap  modules-load.d  plymouth  selinux  terminfo
acpi  cron.yearly  grub.d  legal  mtab  pm  sensors.d  timezone
adduser.conf  crontab  gshadow  libaudit.conf  multipath  polkit-1  sensors3.conf  tmpfiles.d
alternatives  cryptsetup-initramfs  gshadow-  libblockdev  multipath.conf  pollinate  services  ubuntu-advantage
apparmor  crypttab  gss  libibverbs.d  nanorc  sgmllib  ucif.conf
apparmor.d  dbus-1  hdparm.conf  libl-3  needrestart  profile  shadow  udev
appport  debconf.conf  hibagent-config.cfg  locale.alias  netconfig  profile.d  shadow-  udisks2
apt  debian_version  hibinit-config.cfg  locale.conf  netplan  protocols  shells  ufw
bash.bashrc  deluser.conf  hosts  locale.gen  network  python3  skel  update-manager
bash_completion  deymod.d  hosts.allow  localtime  networkd-dispatcher  python3.12  sos  update-motd.d
bindresvport.blacklist  dhcp  hosts.deny  logcheck  networks  rc0.d  ssh  update-notifier
binfmt.d  dhcpcd.conf  hosts.deny  login.defs  nftables.conf  rc1.d  ssl  usb_modeswitch.conf
byobu  dpkg  init.d  logrotate.conf  nginx  rc2.d  subgid  usb_modeswitch.d
ca-certificates  e2scrub.conf  iproute2  lab-release  opt  rc3.d  subuid  wconsole.conf
ca-certificates.conf  ec2_version  inputrc  lvm  os-release  rc4.d  subuid  vim
chrony  envronment  iproute2  machine-id  overlayroot.conf  rc5.d  sudo.conf  vmware-tools
cloud  ethertypes  iscsi  magic  overlayroot.local.conf  rc6.d  sudo.logsrvd.conf  vtrgb
console-setup  fstab  issue  magic.mime  overlayroot.local.conf  rc6.d  sudo.logsrvd.conf  wgetrc
credstore  fuse.conf  issue.net  manpath.config  pam.conf  rmt  sudoers  xattr.conf
credstore.encrypted  fwupd  kernel  mdadm  pam.d  rpc  sudoers.d  xdg
cron.d  gai.conf  landscape  mime.types  passwd  rsyslog.conf  sysctl.conf  zsh_command_not_found
cron.daily  gnutls  ld.so.cache  mke2fs.conf  passwd-  rsyslog.d  sysctl.d
```

```
ubuntu@ip-172-31-11-208:/etc$ cd nginx/
ubuntu@ip-172-31-11-208:/etc/nginx$ ls
conf.d  fastcgi_params  koi-win  modules-available  nginx.conf  scgi_params  sites-enabled  uwsgi_params
fastcgi.conf  koi-utf  mime.types  modules-enabled  proxy_params  sites-available  snippets  win-utf
ubuntu@ip-172-31-11-208:/etc/nginx$ cd sites-available/
ubuntu@ip-172-31-11-208:/etc/nginx/sites-available$ ls
default
ubuntu@ip-172-31-11-208:/etc/nginx/sites-available$ sudo chmod 777 default
ubuntu@ip-172-31-11-208:/etc/nginx/sites-available$
```

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```
root /var/www/html;

# Add index.php to the list if you are using PHP
index index.html index.htm index.nginx-debian.html;

server_name _;

location / {
    proxy_pass http://localhost:4000;
    proxy_http_version 1.1;
    proxy_set_header Upgrade $http_upgrade;
    proxy_set_header connection 'Upgrade';
    proxy_set_header Host $host;
    proxy_cache_bypass $http_upgrade;

    # First attempt to serve request as file, then
    # as directory, then fall back to displaying a 404.
    #try_files $uri $uri/ =404;
}
```

Help

Exit

Write Out

Read File

Where Is

Replace

Cut

Paste

Execute

Justify

Location

Go To Line

Undo

Redo

Set Mark

Copy

To Bracket

Where Was

Previous

Next

```
ubuntu@ip-172-31-11-208:/etc/nginx/sites-available$ sudo chmod 777 default
ubuntu@ip-172-31-11-208:/etc/nginx/sites-available$ nano default
ubuntu@ip-172-31-11-208:/etc/nginx/sites-available$ sudo systemctl restart nginx
ubuntu@ip-172-31-11-208:/etc/nginx/sites-available$
```

Step 4: Go to the incognito tab and paste the instance URL to check the project without entering the PORT.

